

When Tooth-Numbering Detectors Fail Together: A Position-Only Baseline and Closed Gaps on Panoramic X-rays

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Abstract. Tooth-numbering models for panoramic X-rays report high accuracy, but the X-ray format places each tooth in a predictable spot, so part of that accuracy could come from position rather than appearance. We measure this with a position-only baseline: a classifier that sees each tooth's bounding box and no pixels. On 425 uncropped panoramic X-rays (11,602 teeth) with five-fold cross-validation grouped by X-ray, it numbers 78.5% of teeth correctly. Three detectors of different design (YOLOv8x, RT-DETR-l, Faster R-CNN) beat it by 14.3 to 16.7 percentage points, on all 32 tooth numbers, and the gap replicates on a second split. Their errors still follow position. On 1.4% of teeth all three detectors give the same wrong number, and in 91% of these the position-only model gives it too, against under 7% by chance. A sequence of interventions locates these shared errors. Shifting the whole image barely changes the detectors' answers, and masking distant context rarely fixes them. They sit next to a missing tooth five times as often as correct teeth and usually take the missing tooth's number, and they occur where the neighbors have closed the space (AUROC 0.78), while erasing a tooth, which leaves the space open, seldom produces them. We release the per-tooth table and a scorer so new models can be compared with the baseline and on the shared failures.

Keywords: Tooth numbering · Panoramic radiography · Shortcut learning · Error consistency

1 Introduction

Assigning each tooth its FDI number is the first step of most automated dental charting: a caries or a periapical lesion found by a detector is only useful once it is attached to the right tooth. Public benchmarks such as DENTEX [8] score numbering together with detection, and reported accuracies are high.

Panoramic X-rays are a strongly standardized format. The patient bites on a guide, the arch runs left to right across the image, and each tooth number has a typical location. This invites a shortcut [6]: a model can get many numbers right from where a box sits, without reading the tooth. Such a shortcut would be invisible in aggregate accuracy and would fail on the patients who differ from the typical arch, for example after extractions.

We ask how much of tooth numbering position alone explains, and whether detectors fail where position fails. Our contributions: (1) a position-only baseline, trained in seconds on six box features, that numbers 78.5% of teeth correctly under grouped cross-validation, with three detectors beating it by 14 to 17 points on every tooth number; (2) an error analysis showing that the detectors’ shared errors are position-like; (3) four interventions and checks (image shift, context masking, tooth erasure, space closure) that place these errors at closed gaps, next to a missing tooth whose neighbors have moved together so the arch looks complete; and (4) a released per-tooth benchmark with folds, reference predictions and a scorer.

2 Related Work

Tooth numbering. Detection-based numbering on panoramic X-rays is standard, with DENTEX [8] and hierarchical diffusion detectors [9] as recent examples and many DENTEX challenge entries building on YOLO or DETR variants [10,15,4]. Some methods use the arch layout as an explicit prior [1]. Evaluations report accuracy or mAP; we found none that compares a model with what position alone achieves on the same teeth.

Shortcut learning. Models can rely on cues that correlate with the label in training data, from skin markings in dermoscopy [19] to site-specific features in chest X-rays [5] and segmentation [14]; see also [11,3]. These studies show a shortcut by removing or altering the cue. We use the same logic for a cue that is partly legitimate: position does carry information about tooth number.

Error consistency. Whether two models fail on the same inputs beyond chance is measured by error-consistency kappa [7]. We go one step further and compare the wrong answers themselves.

3 Methods

Data and splits. UFBA-425 [2] contains 425 panoramic X-rays (512×512 pixels) with per-tooth masks labeled by FDI number (32 classes, third molars included). Each tooth’s box is the full extent of its mask, giving 11,602 teeth. We use the original X-rays only; an augmented export of this data adds randomly cropped copies, which move teeth in the frame and lowered position-only accuracy by about 8 points in our earlier splits. X-rays are split into five folds stratified by the dataset’s categories; each training fold holds out 15% of its X-rays for model selection, and each X-ray is tested once by models that never saw it. A second split (another seed) shares a test fold with the first for 20.9% of X-rays.

Models. The position-only baseline is a gradient-boosted tree classifier (scikit-learn HistGradientBoosting, default settings) on six features of the labeled box: center x and y , width, height, area and aspect ratio, relative to image size. It never sees pixels. The detectors are YOLOv8x [12], RT-DETR-l [20] and Faster R-CNN [17] with a ResNet-50 FPN backbone [13], all from COCO weights: a

Table 1. Top-1 on 11,602 teeth (split 0, 95% CI) and the gap over the position-only model. F1 counts extra boxes. Split 1: gap on the second split.

	Top-1 (%)	Gap (points)	Numbered F1	Split 1 gap
Position-only	78.5 (76.9–80.0)			
YOLOv8x	95.0 (94.2–95.8)	16.5 (15.1–17.9)	95.0	15.8
RT-DETR-l	95.2 (94.4–95.9)	16.7 (15.3–18.1)	92.3	15.9
Faster R-CNN	92.8 (91.9–93.7)	14.3 (13.0–15.7)	89.6	[pending]

single-stage CNN, a transformer and a two-stage CNN, from two separate code bases. Each is trained for 30 epochs at 640 pixels with horizontal flips off (a flip swaps left and right tooth numbers); Faster R-CNN uses SGD with a learning rate scaled to its batch of 2. The epoch with the best validation fitness ($0.1 \text{ mAP}_{50} + 0.9 \text{ mAP}_{50:95}$) is kept.

Scoring. Detections with confidence at least 0.5 are matched one-to-one to labeled teeth at IoU at least 0.5, highest confidence first, regardless of class. A tooth with no match counts as wrong. Top-1 is the share of labeled teeth matched and given the right number, which puts the detectors and the baseline (which always answers) on the same teeth. We also report numbered F1, which counts extra boxes. CIs are 95% intervals from 10,000 bootstrap draws of X-rays, paired for differences. Each experiment’s decision rule was written and frozen before it was run. A tooth is a *joint failure* when all three detectors and the position-only model are wrong and the detectors give the same answer. The *position match* is the share of joint failures where the position-only model also gives that answer; its chance level comes from shuffling the position-only answers among the joint failures.

Interventions. *Shift*: the whole image is translated by $\pm 10\%$ of its width (about two tooth widths), labels moved to match. *Context masking*: everything outside a tooth’s box widened by k box widths and heights ($k = 0.5, 1, 2$) is blacked out. *Missing neighbor*: an adjacent arch position has no labeled tooth. *Tooth erasure*: for up to three teeth per X-ray with both neighbors present, the tooth’s mask (dilated 3 px) is inpainted [18], and we ask whether a correctly numbered neighbor now takes the erased tooth’s number; controls are the teeth two places away and a sham inpaint over bone. *Space closure*: for each missing tooth with both neighbors present, S is the shortest distance between the neighbors’ masks over the missing tooth’s median width (0 when they touch).

4 Results

Position explains most; detectors add 14 to 17 points. Table 1. Position alone numbers more than three in four teeth. Each detector beats it on all 32 tooth numbers; the per-number gap is smallest at the third molars (4.3 to 5.8 points) and largest at the lower central incisors (up to 25.5). Given the detectors’ own predicted boxes instead of labeled ones, the baseline does slightly better

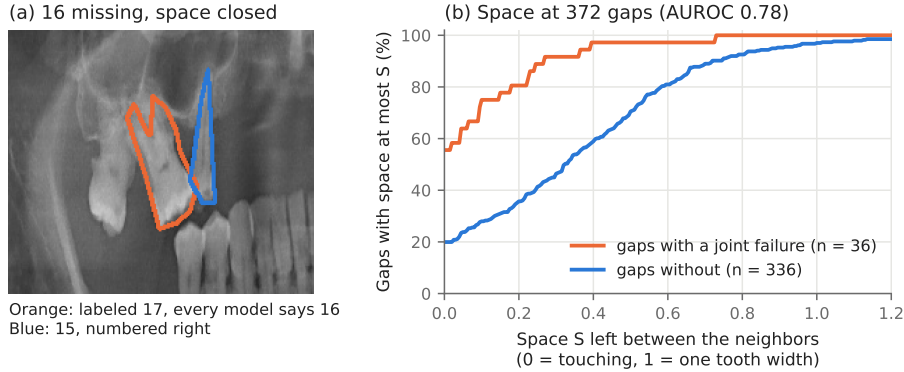


Fig. 1. (a) A closed gap: tooth 16 is missing and its neighbors' outlines touch. All three detectors and the position-only model number tooth 17 as 16. (b) Share of the 372 gaps with both neighbors present whose space is at most S , for gaps with and without a joint failure.

(79.0 to 80.2%) and the gap is 14.5 to 16.2 points, so labeled boxes do not flatter it. On the second split the gaps fall inside the first split's CIs. YOLOv8x and RT-DETR-l tie on top-1 (difference 0.17 against a minimum detectable difference of 0.50), but RT-DETR-l leaves about 1.9 unmatched boxes per X-ray against 0.4, which lowers its F1 by 2.7 points.

Shared errors follow position. 161 teeth (1.4%) are joint failures. In 91.3% of them the position-only model gives the same wrong number, against a permutation 95th percentile of 6.8%, and 94.4% of the shared wrong numbers are one place away in the same quadrant. The detectors are about as confident on these teeth as on correct ones (AUROC 0.53 to 0.66), so a confidence threshold does not find them, and neither does a rule based on the detectors' second choices. About 10% of joint failures may be label errors by confident learning [16]; without them the position match is 91.7%.

Not absolute position, not distant context. Shifting the image by two tooth widths, which leaves the position-only model right on under 9% of teeth, changes the detectors' top-1 by under 0.3 points. Masking everything beyond one box width costs Faster R-CNN and RT-DETR-l about 26 points on correctly numbered teeth (YOLOv8x 9.6), but among detected joint failures 74 to 93% keep the shared wrong number.

Next to missing teeth. 44.7% of joint failures have a missing neighbor, against 8.5% of teeth all three detectors get right (ratio 5.3, CI 4.2 to 6.6). 67 of the 72 joint failures next to a gap (93%) take the missing tooth's number. Teeth that only the position-only model gets wrong have a missing neighbor 10.5% of the time: gaps are not where position fails in general, but where position and the detectors fail together.

At closed gaps. Fig. 1. Across 372 missing teeth with both neighbors present, the space left between the neighbors separates gaps with a joint failure

from gaps without one (AUROC 0.78, CI 0.70 to 0.84; median S 0.00 against 0.32). 56% of these errors sit at gaps whose neighbors touch, against 20% of other gaps. Each detector’s own errors track closure as well (AUROC 0.74 to 0.79), more than the position-only model’s (0.68).

An open gap is not enough. Erasing a tooth makes a correctly numbered neighbor take its number in 2.0% (YOLOv8x), 4.4% (RT-DETR-l) and 5.7% (Faster R-CNN) of cases, against 0 to 0.2% for both controls. But all three detectors do it to the same neighbor in only 4 of 2,260 cases (0.18%).

5 Discussion

The detectors read teeth: they do not number by position in the frame, and they beat a position-only model on every tooth. Yet their rare shared errors are the position-only model’s errors, and our results show where they happen. When a tooth is missing and its neighbors have moved together, the arch shows no gap. Counting along it gives the tooth next to the closed space the missing tooth’s number, and that tooth also sits roughly where the missing one would. Appearance, context and position then point the same way, and every model follows. An inpainted gap leaves a full-width space and does not create this agreement.

Two practical points follow. A position-only baseline is cheap and should be reported with tooth-numbering results: on standardized panoramic X-rays it sets a high floor, and the margin above it is what a model contributes. And the errors that different models share are not found by confidence; a review rule from outside the model works better: check the numbering next to every missing tooth, especially where the space has closed. Here such a rule catches 97% of the shared errors at gaps but flags many gaps without one (precision 13%), since most gaps are partly closed.

Limitations. One dataset from one source, a single annotation per tooth and no demographic data; we did not validate on an external dataset such as DENTEX. The closure result is correlational, and masks on a 2D projection can touch when the teeth do not. Third-molar and longer gaps, which hold most missing positions, are outside the closure analysis. A second split replicates the main numbers but changes split and training randomness together. About 10% of joint failures may be label errors.

6 Conclusion

On uncropped panoramic X-rays, position alone numbers 78.5% of teeth and detectors add 14 to 17 points. The few errors they share follow position and concentrate next to missing teeth whose neighbors have closed the space. We release the per-tooth table and scorer so that new models can be measured against the baseline and on these shared failures.

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